DON BOSCO
INSTITUTE OF TECHNOLOGY
Bengaluru, Karnataka

AWS STUDENT
BUILDER GROUP
DBIT

AFTERHOURS 1.0

CODE BEYOND THE CLOCK

THEME 05: EDGE INTELLIGENCE & OFFLINE-FIRST AI

VOICEKIRANA

Offline Voice-Based Stock, Credit Ledger & Instant Bill Generator

"Speak vernacular entries. Generate instant bills, track udhaar, and prevent stockouts — 100% offline."

TEAM DETAILS

Team Name: AI Avengers

Track: THEME 05: EDGE INTELLIGENCE & OFFLINE-FIRST AI

Don Bosco Institute of Technology (DBIT)

TEAM MEMBERS

- **Raj Aanada**
- **Yashma C P**
- **Likith M**
- **Yashwanth K R**

Problem Statement & Its Relevance

The Small Merchant Digital Divide & Manual Ledger Chaos. *Kirana owners lose hours and ₹1,25,000+ monthly due to manual paper bookkeeping during busy counter rushes.*

COUNTER FRICTION

THE PAPER BAHI-KHATA TRAP

- **Typing is Too Slow:** During peak evening rush, typing on mobile apps or desktop POS keyboards causes long customer lines.
- **Forgotten Credit (Udhaar):** Writing credit entries on loose paper slips leads to forgotten balances and disputes.
- **Zero Inventory Visibility:** Shopkeepers run out of staples (sugar, oil, atta) mid-day without warning.

CURRENT GAPS

WHY EXISTING TOOLS FAIL

- **Cloud Apps Crash Offline:** Khatabook & Vyapar freeze completely when internet connectivity drops.
- **Desktop POS is Prohibitive:** Hardware costs ₹15,000+ with steep learning curves for semi-literate clerks.
- **No Instant Bill Generation:** Voice apps only record memos; they cannot compute taxes or print thermal receipts.

ECONOMIC LOSS

WHY DOES IT MATTER?

- **₹1,25,000 Monthly Loss:** Average unrecovered credit and unbilled items per small rural/urban grocery store.
- **Spotty Connectivity:** Tier-2/3 towns and basement stores suffer frequent cellular drops; cloud apps freeze.
- **Language Exclusion:** Over 80% of merchants cannot navigate complex English POS interfaces.

EMPIRICAL SCALE

GROUND REALITY IN NUMBERS

- **13+ Million Kiranas in India:** Drive 88% of FMCG retail sales across India, yet 92% operate on manual paper notebooks.
- **2.5 Hours Daily Lost:** Spent matching customer ledgers and stock counts after shop closure every single night.
- **14% Stockouts on Staples:** Loss of high-margin walk-in sales when milk, oil, and flour unexpectedly run dry.

13M+

Kirana Stores in India

88% of FMCG retail runs on offline paper ledgers

₹1.25L

Monthly Credit at Risk

Unreconciled udhaar lost per store annually

2.5 Hrs

Daily Bookkeeping Saved

Time reclaimed by hands-free voice billing

Proposed Solution & Core Innovation

Voice-to-Ledger-to-Bill: 100% Offline Edge Retail Operating System. *Speak naturally in vernacular dialect; VoiceKirana generates instant bills, updates udhaar, and alerts low stock.*

VERNACULAR SPEECH ENGINE

Offline Edge Speech-to-Text

Understands regional dialects (Hindi, Kannada, Hinglish) using quantized Vosk ASR running 100% locally with zero cloud latency.

SLOT-FILLING BILL GENERATOR

Instant Itemized Thermal Receipt

Parses "2 kg sugar Ramesh udhaar" into structured items, computes total with tax, prints thermal bill, and shows customer UPI QR.

DOUBLE-ENTRY CREDIT KHATA

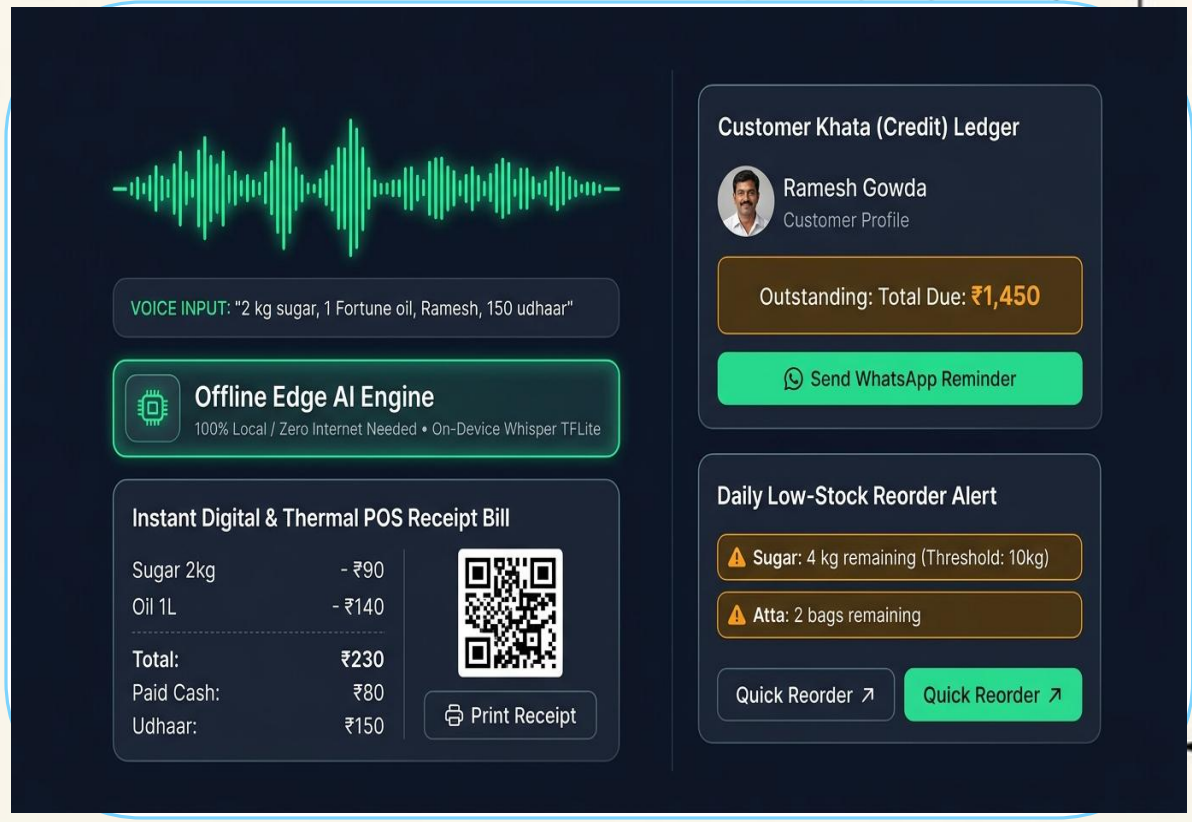
Zero-Dispute Customer Ledger

Maintains ACID-compliant SQLite credit balances per customer with instant SMS/WhatsApp summary sync when connectivity restores.

PREDICTIVE STOCK SENTINEL

End-of-Day Low Stock Alerts

Automatically decrements inventory with every voice entry; generates wholesale reorder shopping lists before shelves empty.



< 300ms SPEECH RECOGNITION
Sub-second quantized on-device inference with zero cellular or Wi-Fi reliance.

INSTANT THERMAL & UPI RECEIPT
Direct Bluetooth thermal printer output + QR code for customer UPI payment.

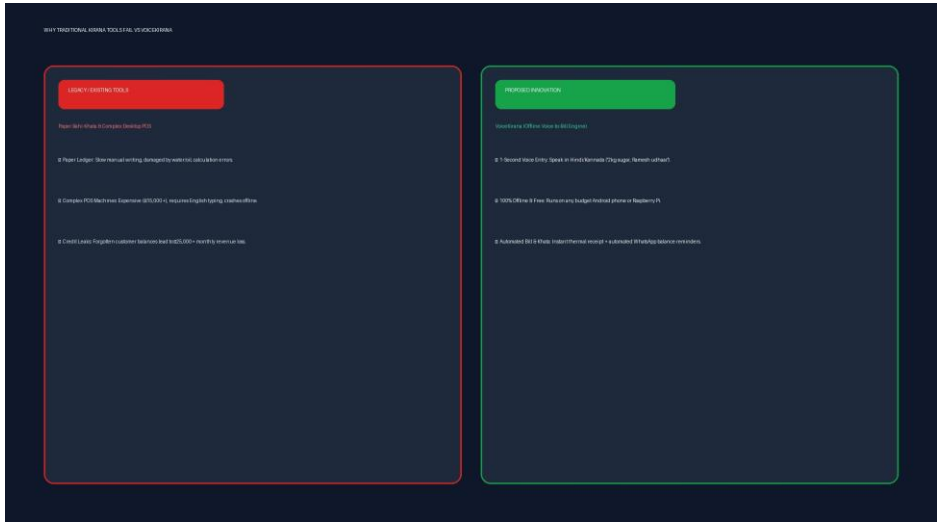
100% PRIVATE LOCAL STORAGE
Merchant financial data and customer phone numbers never leave the device.

Existing Solutions & Differentiation

Why Paper Khatas and Desktop POS Systems Fail in Kirana Stores. *Comparing VoiceKirana against traditional paper records and heavy cloud billing software.*

TRADITIONAL BOTTLENECK

LEGACY SOLUTIONS & DAILY FAILURES



- **Paper Bahi-Khata: Manual writing causes counter delays, damaged pages, and lost debts.**
- **Khatabook / Vyapar: Cloud-dependent; typing required; unusable when internet drops.**
- **Desktop POS: Expensive (₹15,000+), bulky hardware, complex English navigation.**

FEATURE MATRIX

BENCHMARK: VOICEKIRANA VS ALTERNATIVES

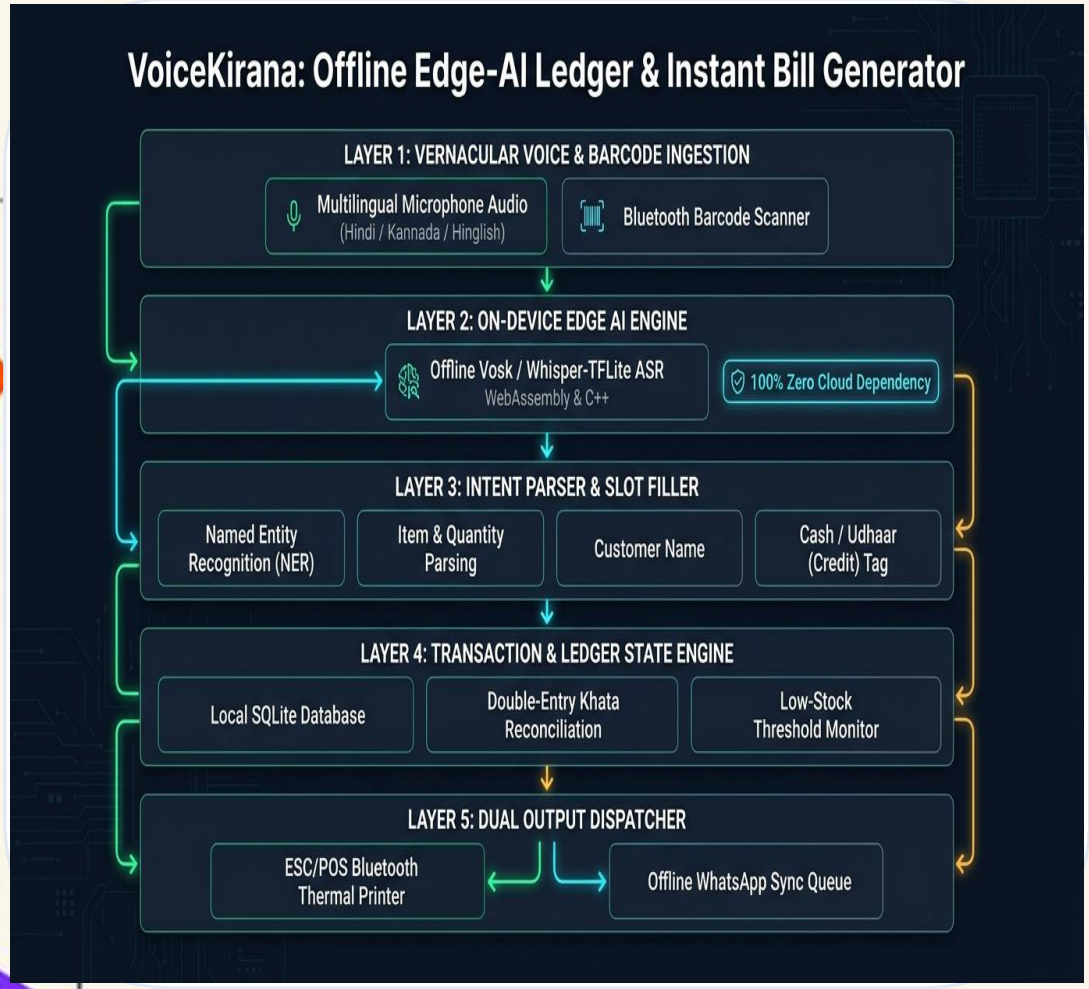
Capabilities	Paper Khata	Khatabook	Desktop POS	VoiceKirana
Hands-Free Vernacular Voice	Manual	Typing Only	English Keys	**Sub-300ms Audio**
100% Offline Functionality	Offline Paper	Fails Offline	Partial Offline	**Pure Edge SQLite**
Instant Bill Generation	None	Manual Print	Bulky Hardware	**1-Sec Voice Bill**
Credit (Udhaar) Khata	Error-Prone	Manual Type	Clunky	**Voice-Linked Khata**
End-of-Day Low-Stock Alerts	None	None	Manual Setup	**Auto-Reorder List**

CORE ARCHITECTURAL ADVANTAGE

- **Zero Training Curve:** Merchants speak the exact same colloquial words they say to customers.
- **Hardware Democratization:** Runs on any existing ₹6,000 Android smartphone with zero cloud hosting charges.

Technical Architecture & System Flow

5-Layer Disconnected Edge Architecture with Real-Time Audio Pipeline. *Ultra-compact offline inference stack: Audio Stream -> Quantized Vosk ASR -> Slot Filler -> SQLite -> ESC/POS.*



LAYER 1: AUDIO INGESTION

Local Microphone Stream & Noise Filter

Pulls 16kHz mono PCM stream via PyAudio; applies spectral subtraction to eliminate grocery shop background chatter.

LAYER 2: QUANTIZED EDGE ASR

Vosk / Kaldi Neural Acoustic Model

Runs 35MB 8-bit quantized acoustic model on edge CPU; outputs phoneme lattices and greedy-decoded text in <300ms.

LAYER 3: SLOT-FILLING NLP

Deterministic Rule & Entity Parser

Extracts Item Name, Quantity, Unit (kg/g/pkt), Price, and Customer Name via high-speed regex & fuzzy catalog matching.

LAYER 4: ATOMIC LEDGER STORE

Embedded SQLite ACID Database

Updates customer khata, decrements stock inventory, and logs transactions in WAL-mode SQLite database with zero cloud sync.

ER 5: THERMAL & DISPLAY DISPA

ESC/POS & UPI Payment Terminal

Generates binary ESC/POS print commands for Bluetooth thermal printers and renders dynamic UPI QR code on merchant screen.

Feasibility, 24-Hour Roadmap & Societal Impact

Achievable 24-Hour Hackathon Delivery, Live Demos & Deep Bharat Reach. *Engineered specifically for live judging: speak vernacular items into mic and watch the thermal bill print instantly.*

LIVE PROTOTYPE

LIVE HACKATHON DEMO PLAN

- **Step 1 (Cash Sale):** Speak "2 kg atta, 1 litre mustard oil" -> Watch instant bill appear in 1.2 seconds.
- **Step 2 (Credit Khata):** Speak "1 kg basmati rice Suresh udhaar" -> Customer khata ledger debited automatically.
- **Step 3 (Low Stock):** Inventory falls below 5kg threshold -> Instant amber low-stock alert triggers on screen.
- **Step 4 (Thermal Print):** One-click print dispatches itemized slip to simulated ESC/POS thermal printer.

TARGET BENCHMARKS

EVALUATION TARGETS

- **< 300ms ASR Latency:** Sub-second speech-to-text response on local CPU hardware.
- **> 95% Entity Extraction:** Accurate extraction of quantity, product name, and price.
- **0 KB Cloud Dependence:** Complete standalone execution with Wi-Fi/cellular turned OFF.
- **< 35MB Model Footprint:** Ultra-lightweight quantized neural model runs on 1GB RAM.

STAKEHOLDERS

WHO BENEFITS?

- **Kirana & Provision Stores:** Instant hands-free billing during peak morning and evening rush hours.
- **Vegetable & Fruit Vendors:** Rapid itemized calculation without paper notes or mental arithmetic errors.
- **Rural Customers:** Transparent itemized thermal printed receipts with clear credit records.
- **FMCG Wholesalers:** Timely stock replenishment orders generated automatically every night.

24-HOUR SPRINT TIMELINE

RIGOROUS 24-HOUR HACKATHON EXECUTION ROADMAP

PHASE 1: 00:00 — 06:00
OFFLINE ASR & MODEL SETUP
- Vosk quantized Indian English/Hindi model setup.
- Local audio capture stream (PyAudio / SoundDevice).
- Word error rate benchmarking on edge CPU.

PHASE 2: 06:00 — 12:00
NLP & SLOT-FILLING ENGINE
- Entity regex parser (item, quantity, price, customer).
- Currency & unit normalizer (kg, g, packet, litre).
- Khata credit vs. cash classifier.

PHASE 3: 12:00 — 18:00
SQLITE LEDGER & THERMAL BILL
- Schema design (products, customers, transactions).
- Instant thermal receipt generator + UPI QR code.
- Automated low-stock trigger logic.

PHASE 4: 18:00 — 24:00
UI POLISH & LIVE MIC REHEARSAL
- Streamlit merchant dashboard with live audio waveform.
- Multi-dialect testing (Hindi, Kannada, Hinglish).
- Final live demo script rehearsal with 5 rapid entries.

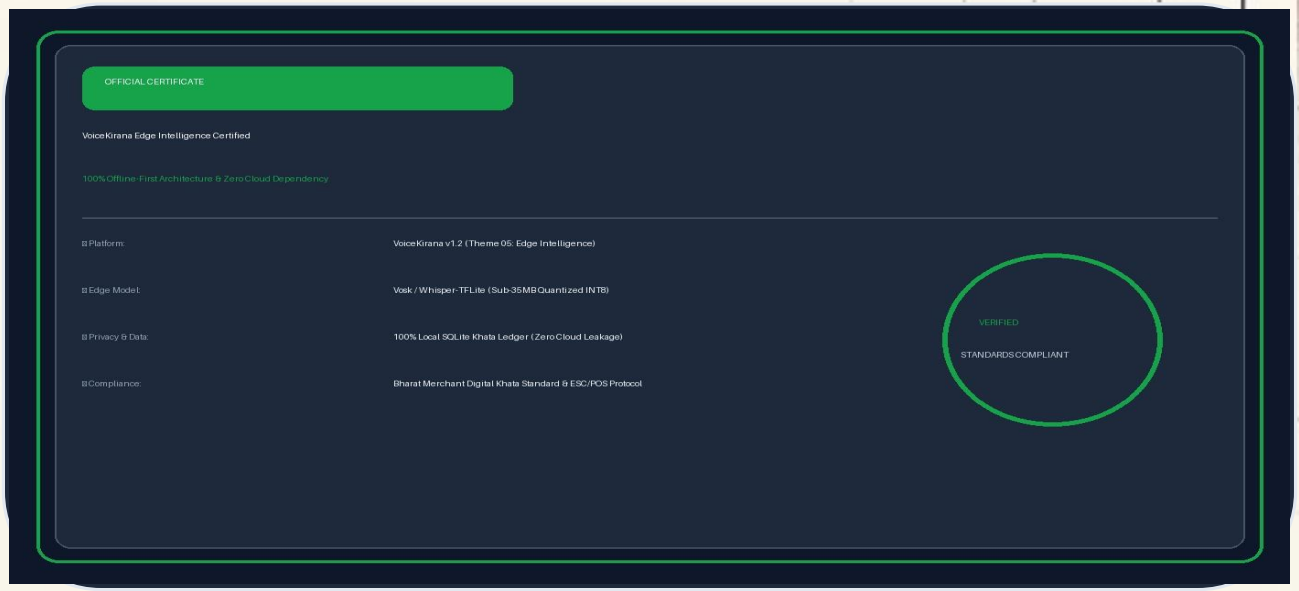
Conclusion & References

Dignity, Speed, and Financial Health for Bharat's Backbone Retailers. *"Bringing edge intelligence to the kirana counter without requiring English or internet."*

CORE SUMMARY

KEY TAKEAWAYS

- **Zero Typing Barrier:** Shopkeepers simply speak naturally; VoiceKirana generates itemized bills in under 3 seconds.
- **100% Offline Autonomy:** Complete immunity from cellular outages; merchant records stay private on the device.
- **Credit & Stock Clarity:** Eliminates ₹1,25,000 monthly credit leaks and alerts merchants before staples run out of stock.
- **Zero Hardware Cost:** Runs on any budget Android phone or existing merchant tablet with zero subscription fees.



TECHNICAL EVIDENCE

VERIFIABLE REFERENCES & EDGE AI BENCHMARKS

1. **Vosk Offline ASR Architecture:** Kaldi-Based Lightweight Speech Recognition for Embedded Devices (AlphaCephei).
2. **TensorFlow Lite Quantization:** Quantized Neural Network Inference for Microcontrollers and Edge Hardware.
3. **SQLite Embedded Database:** ACID Compliance and Crash-Proof Storage in Disconnected Environments.
4. **National Retail Federation Report:** Digitization and Informal Credit Dynamics in Indian Unorganized Retail.
5. **ESC/POS Protocol Standard:** Binary Command Set for Point of Sale Thermal Printing Systems.
6. **NPCI UPI Offline Specification:** Dynamic QR Code Generation for Contactless Merchant Payments.